Methods & Overloading: Student.java

Student.java (Part 1):

```java
// Student.java
public class Student {
    public String firstName;
    public String lastName;
    public int studentId;

    public Student() {
        // Default constructor from previous assignment.
    }

    // Method to perform a simple calculation.
    public double calculate(double num1, double num2, char operator) {
        switch (operator) {
            case '+':
                return num1 + num2;
            case '-':
                return num1 - num2;
            case '*':
                return num1 * num2;
            case '/':
                if (num2 != 0) {
                    return num1 / num2;
                } else {
                    System.out.println("Error: Division by zero is not
allowed.");
                    return 0; // Return 0 or throw an exception.
                }
            default:
                System.out.println("Error: Invalid operator.");
                return 0;
        }
    }
}
```

Student.java (Part 2):

```java
// Student.java
import java.util.regex.Matcher;
import java.util.regex.Pattern;

public class Student {
    public String firstName;
    public String lastName;
    public int studentId;

    // Overloaded constructors
    // Default constructor
    public Student() {
        this("Unknown", "Unknown", 0);
    }
    // Constructor with all details
    public Student(String firstName, String lastName, int studentId) {
        this.firstName = firstName;
        this.lastName = lastName;
        this.studentId = studentId;
    }
    // Constructor with first and last name
    public Student(String firstName, String lastName) {
        this(firstName, lastName, 0);
    }
    // Constructor with student ID only
    public Student(int studentId) {
        this("Unknown", "Unknown", studentId);
    }

    // Overloaded calculate method (two numbers and a character operator)
    public double calculate(double num1, double num2, char operator) {
        switch (operator) {
            case '+':
                return num1 + num2;
            case '-':
                return num1 - num2;
            case '*':
                return num1 * num2;
            case '/':
                if (num2 != 0) {
                    return num1 / num2;
```

```java
                } else {
                    System.out.println("Error: Division by zero is not 
allowed.");
                    return 0;
                }
            default:
                System.out.println("Error: Invalid operator.");
                return 0;
        }
    }

    // Overloaded calculate method (equation as a string)
    public double calculate(String equation) {
        // Use a regular expression to parse the equation string
        Pattern pattern = Pattern.compile("(\\d+)\\s*([+\\-*/])\\s*(\\d+)");
        Matcher matcher = pattern.matcher(equation);

        if (matcher.find()) {
            double num1 = Double.parseDouble(matcher.group(1));
            char operator = matcher.group(2).charAt(0);
            double num2 = Double.parseDouble(matcher.group(3));
            return calculate(num1, num2, operator);
        } else {
            System.out.println("Error: Invalid equation format. Please use 
'Num1 Op Num2'.");
            return 0;
        }
    }
}
```